PINTAR.ai – Platform for INclusive Trade & AI Resilience

Case Study 8: AI for Inclusive MSMEs Growth

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1.0 Introduction

1.1 Background of Case Study

This project is developed in response supervised under Case Study 8: AI for Inclusive Growth, which highlights a critical economic paradox within the ASEAN community: the widening disparity between MSME business prevalence and their actual macroeconomic contribution.

While Micro, Small, and Medium Enterprises (MSMEs) form the backbone of ASEAN economies—comprising 97% of all businesses and over 85% of regional employment—their contribution to national GDP (30–53%) and export volume (10–30%) remains disproportionately low. This stagnation is largely driven by financial invisibility, limited digital literacy, and the overwhelming complexity of cross-border trade compliance.

To counteract this phenomenon, PINTAR.ai (Platform for INclusive Trade & AI Resilience) was engineered. Designed to empower both banked and unbanked enterprises, PINTAR.ai bridges the gap between informal trade practices and global financial standards. For the scope of this initial deployment and case study, PINTAR.ai focuses its target segment on MSMEs operating within Malaysia, particularly serving underserved and rural demographics.

1.2 Chosen Sustainable Development Goals

PINTAR.ai's architecture and mission are deeply aligned with the United Nations Sustainable Development Goals (SDGs), specifically addressing three core pillars:

SDG 8: Decent Work and Economic Growth At its core, PINTAR.ai is an economic equalizer. By providing enterprises in marginalized ASEAN communities with enterprise-grade tools—such as automated proxy credit scoring and cross-border intelligence—the platform systematically dismantles the digital and economic disparities that prevent MSMEs from scaling, accessing formal capital, and participating equitably in the global economy.

SDG 9: Industry, Innovation & Infrastructure PINTAR.ai establishes a new paradigm for inclusive digital infrastructure. Through the creative fusion of multi-agent AI pipelines, predictive analytics, and localized automation (such as OCR for informal trade receipts), the platform provides MSMEs with advanced technological capabilities. This equips traditional businesses to seamlessly integrate AI to forecast business prospects and innovate their operational models.

SDG 10: Reduced Inequalities PINTAR.ai directly addresses the systemic inequalities faced by unbanked or predominantly offline MSMEs. By translating complex regulatory environments into accessible insights and digitalizing conventional financing benchmarks without requiring formal bookkeeping literacy, the platform reduces the economic and digital divide between top-tier corporations and grassroots entrepreneurs in the region.

1.3 Problem Statement

No.	Problem (Simplified)	Elaboration/Impact	Key Inference
1.	**Low awareness and knowledge of digital technologies**	MSMEs lack information on digital tools' benefits, face language barriers (most content in English), and rely on word-of-mouth, blocking the first step toward adoption and productivity gains.	Only basic tools (email/social media) are used; advanced applications (analytics, ERP) remain rare, capping GDP/export shares and keeping businesses local and unscaled (Spire Research and Consulting, 2018).
2.	**Barriers to digital trade participation and cross-border operations**	Uneven broadband costs/speeds, data-localisation rules, inconsistent de-minimis thresholds, and weak trade facilitation raise compliance and logistics costs for small shipments.	Webpage adoption as low as 13% (Cambodia manufacturing); data restrictions reduce competitiveness. MSMEs stay domestic-focused, explaining the 85%+ employment share vs. ~18% export share gap (López González, J., 2019).
3.	**Severely restricted access to formal financing**	Banks demand collateral/credit history that most MSMEs lack; high perceived risk and underdeveloped non-bank channels result in low lending penetration and high rejection rates.	MSME bank loans average only ~10–20% of total lending or GDP in many AMS, starving investment in productivity/tech and keeping output and export contributions low (Asian Development Bank., 2023).
4.	**Critical gaps in digital skills and human capital**	Education systems lag behind rapid tech change; low STEM/lifelong-learning focus and industry mismatch mean owners and workers cannot operate, maintain, or upgrade digital tools.	Skills shortages cited as a top barrier to adoption across AMS, preventing productivity jumps and e-commerce/export scaling, reinforcing the suboptimal GDP contribution (World Bank., 2019).
5.	**Access-to-finance and market-access bottlenecks for digitalisation**	66%+ of firms (e.g., Cambodia) cite finance as a major hurdle; market access remains domestic-only due	MSMEs contribute 50–95% employment but only 30–53% GDP; finance/market gaps keep them informal/unbanked and unable to

	to weak digital platforms, logistics, and standards in many AMS.	reach global customers/investors (ASEAN Secretariat., 2023).

1.4 Objectives & Proposed Solutions

Objectives:

1. **Cross-Border Readiness:** To prepare ambitious MSMEs for cross-border trade and international business expansion.
2. **Financial Visibility:** To provide an instant analytical lookup of MSMEs' eligibility for bank loans and formal credit.
3. **Inclusive Digitization:** To digitalize conventional financing documentation for unbanked or predominantly offline businesses.
4. **Marketing Empowerment:** To empower MSMEs with the ability to seamlessly integrate digital solutions into their business promotion and ad creation.

Proposed Solutions:

1. **Pre-Screening Dashboard:** Addresses Objective 2 by ingesting user financial data and outputting a standardized Proxy Credit Score (300-850) using the Qwen-3 VL engine.
2. **Shadow Ledger:** Addresses Objective 3 by utilizing localized OCR (featuring Pua Kumbu motif recognition) to digitize informal, offline trade receipts into structured ledger data.
3. **VernStudio (Snap & Sell 2.0):** Addresses Objective 4 by instantly generating 8-second vertical video advertisements localized for ASEAN cultural contexts, allowing quick deployment of products (like Bario Rice).
4. **Compliance Sentinel:** Addresses Objective 1 by providing interactive, real-time regulatory mapping and export intelligence to guide MSMEs through complex international trade compliance.

2.0 Review of Existing Solutions

A. Receipt/Transaction Digitization — Shadow Ledger

Competitor	Approach	Limitation
Wave / QuickBooks	Manual entry or bank feed sync	Requires bank API integration; no support for informal cash/WhatsApp receipts
Autocount (MY)	Desktop accounting software	Needs training; no mobile OCR; targets formalized businesses only
StoreHub POS	Point-of-sale system with analytics	Requires hardware; no receipt photo capture; can't process WhatsApp screenshots
Manual bookkeeping	Paper ledger / Excel	70% of Sarawak MSMEs still operate this way — no AI, no structure

Gap PINTAR.ai fills:

- **Photo → structured ledger.** Lack of dedicated MSME tool in Malaysia converts a receipt photo or WhatsApp chat screenshot into a classified, sentiment-scored financial entry using OCR + LLM.
- **Vernacular support.** Processes Manglish, Sarawak Malay, Iban, and Foochow — languages that zero existing tools support.

- **Intent classification.** Automatically categorizes entries as PAYMENT_IN, ORDER_IN, or COMPLAINT — something no competitor offers for informal transactions.

B. Product Listing Generation — Snap & Sell

Competitor	Approach	Limitation
Shopee Seller Centre	Manual text entry + template	Requires typing proficiency; no voice input; auto-translation limited to BM compliance
Lazada Seller Centre	Manual listing + attribute filling	500-product limit for new sellers; no multilingual generation beyond basic translation
8Web Malaysia	Human copywriting service	Expensive; slow turnaround; not scalable for micro-enterprises
ChatGPT / generic AI	Prompt-based text generation	No product photo analysis; no background removal; no Borneo language ASR
Sarawak Digital Mall	TikTok Shop + MINTRED collaboration	Sales channel only; no listing generation tools; no voice-to-text pipeline

Gap PINTAR.ai fills:

- **Voice-first, photo-first.** Rural artisans speak their product description in Iban/Foochow and snap a photo — zero typing required. No competitor does this.
- **Malaysia-AI ASR.** Uses mesolitica/whisper fine-tuned for Malaysian languages — not available in any e-commerce platform's built-in tools.
- **Multimodal listing.** Kimi-K2.5 sees the product photo AND the transcript to generate contextually rich, culturally authentic bilingual (EN/ZH) listings. Shopee/Lazada only offer basic machine translation.
- **Background removal + enhancement.** Automated product isolation (rembg/U^2-Net) built directly into the pipeline — no separate tool needed.

C. Cross-Border Compliance — Customs Navigator

Competitor	Approach	Limitation
MATRADE MDT Platform	Business matchmaking + market intelligence	No product-specific compliance check; no traffic-light system; designed for established exporters
MITI Trade Portal	Static regulation documents	Manual lookup; no AI interpretation; overwhelms micro-entrepreneurs
XTransfer (BNM-approved 2026)	Cross-border payment processing	Payments only — no regulatory compliance guidance whatsoever
Mastercard Global Commerce Suite	Multi-currency payment simplification	Payment infrastructure; zero compliance or customs advisory
Wallex / M-DAQ	FX conversion and cross-border transfers	Currency/remittance focused; no export guidance
Customs brokers	Human advisory	RM 500–2,000 per consultation; inaccessible to micro-MSMEs

Gap PINTAR.ai fills:

- **Instant, AI-powered compliance verdicts.** Enter "peppercorn → Singapore" and get a traffic-light assessment (🟢🟡🔴) with HS code estimation, duty rates, and step-by-step requirements. No tool in ASEAN does this for MSMEs.
- **RAG-grounded on DEFA documents.** The Customs Navigator uses vector search over official ASEAN DEFA regulatory documents — not hallucinated LLM knowledge, but retrieval-augmented generation with cited sources.
- **Multi-data fusion.** Combines regulatory context (DEFA RAG), live trade barriers (WTO SPS/TBT), logistics data (freight index, transit days), and the MSME's own financial position into a single strategic assessment. No competitor fuses these 3 data streams.
- **Accessibility.** Free, instant, mobile-friendly — versus RM 500+ per customs broker consultation.

D. Financial Eligibility & Credit Scoring — Dashboard & Bank Analyzer

Competitor	Approach	Limitation
CCRIS / CTOS	Traditional credit bureau scoring	Requires formal credit history; excludes informal/cash-based MSMEs
Funding Societies (MY)	P2P lending with manual document review	Requires audited financials; slow human underwriting
Begini	Alternative credit scoring (digital footprint)	Mobile/behavioral data focus; no bank statement AI analysis
CRIF Malaysia	Alternative data credit scoring	Enterprise-grade B2B tool; not consumer/MSME-facing
Finory.tech	AI bank statement analyzer	Generic tool; no Malaysian MSME context; no KPI benchmarking against BNM tiers

Gap PINTAR.ai fills:

- **PDF → full audit in seconds.** Upload a bank statement PDF and get 6 KPIs (Net Cash Flow, DSCR, Expense Ratio, Revenue Consistency, Avg Balance, Overdrafts), a proxy credit score (300–850), risk classification, and an AI-generated strategic summary.
- **BNM-tier benchmarking.** Automatically classifies MSMEs by BNM's size categories (micro <RM300k, small <RM15M, medium) and benchmarks KPIs accordingly.
- **What-if simulation.** Interactive sliders let MSMEs simulate "What if I increase revenue by RM5,000?" or "What if I cut expenses by RM2,000?" and instantly see how their eligibility score changes.
- **CCRIS/CTOS proxy.** For unbanked/thin-file MSMEs, generates a proxy credit score from statement patterns — no credit bureau history needed.

E. Predictive Analytics — Neuro-Symbolic Engine

Competitor	Approach	Limitation
StoreHub Analytics	POS-based sales reporting	Backward-looking only; no predictive; no macro-data fusion
Xero / MYOB	Cloud accounting with dashboards	Requires structured data input; no ASEAN macro-economic context
Generic BI tools	Tableau, Power BI	Requires data literacy; no MSME-specific insights; overkill for micro-enterprises

Gap PINTAR.ai fills:

- **Micro + macro data fusion.** Combines the MSME's own Convex ledger data (micro) with ASEAN economic indicators (macro) through deterministic math + LLM strategic narrative.
- **Export compass.** Ranks target markets by friction score, considering the MSME's actual product, financial position, and destination-specific barriers.
- **Liquidity warnings.** Real-time 7-day income-to-expense ratio monitoring with severity alerts (green/yellow/red).

2.3 Methodology

PINTAR.ai lives up by its name, serving as a platform that empowers the ability of foreseeing, managing and expanding their businesses digitally while implementing "pre-screening" before they made any decisions where their current business may, or may not be able to achieve. The technical stack of PINTAR.ai is as described in **Table 3.1**.

No.	Technical Stack	Specifications
1	Frontend (Client)	TypeScript, Next.js 15, React, Tailwind CSS
2	Backend (Server)	Python, FastAPI
3	Database	Convex (Reactive DBMS)
4	OAuth/User Registration & Login	Clerk (OAuth via Github/Gmail)
5	Infrastructure/Deployment	Vercel.
6	Development Tools	Git, VSCode, Google Antigravity, Twillio, Ngrok.

***Table 3.1**: Specifications of technical stack of PINTAR.ai.*

The system architecture of PINTAR.ai adopts a modern, decoupled hybrid framework. The client-side operations are handled by a serverless Next.js 15 frontend deployed on Vercel, which communicates directly with a real-time reactive database (Convex) and an Identity Provider (Clerk) for authentication. Intensive computational tasks—such as Natural Language Processing (NLP), asynchronous audio transcription, vector embeddings, and image processing—are offloaded to a specialized Python FastAPI backend. The flow of data between the user interfaces, the reactive database, and the AI models is visualized in Figure 3.1.

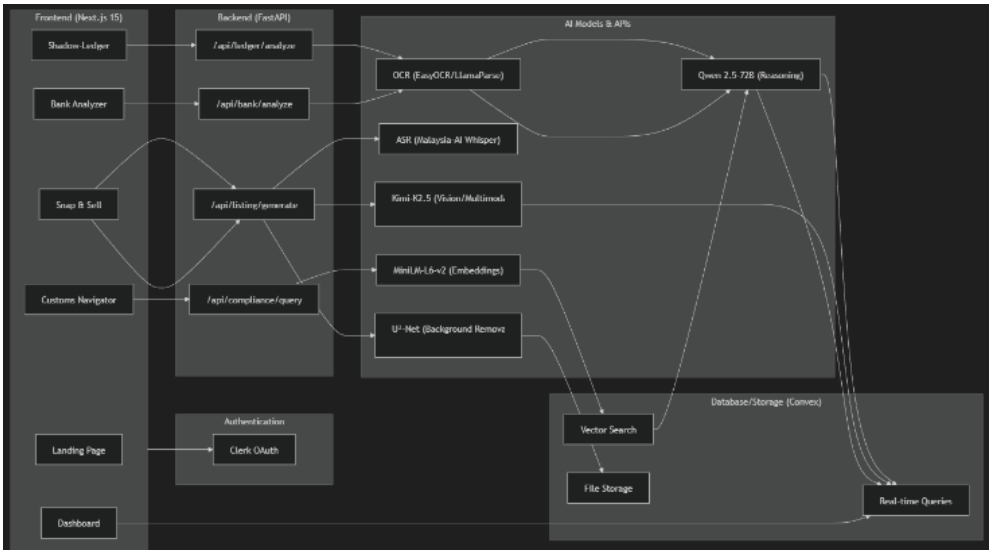


Figure 3.1: System Architecture Diagram of PINTAR.ai.

1. **Shadow Ledger (Receipt Digitization)**

This module converts informal receipt photos and WhatsApp chat screenshots into structured financial ledger entries, targeting MSMEs that do not maintain formal accounting records.

Execution Pipeline:

i. **User Upload:** The user uploads a receipt or transaction screenshot via the frontend interface.

ii. **Preprocessing:** The backend applies image enhancements (grayscale conversion, contrast elevation, sharpening) to optimize the image for extraction.

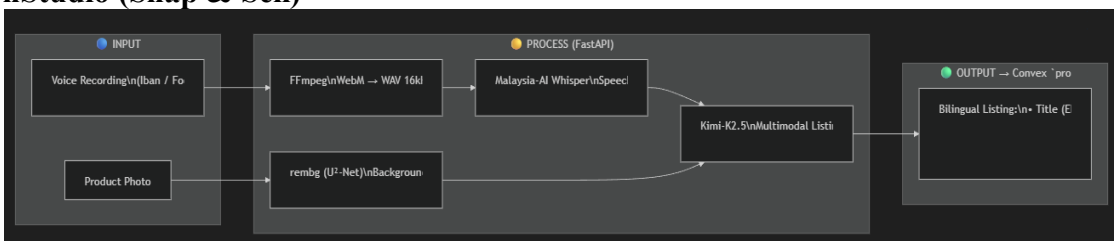
iii. **OCR Extraction:** Optical Character Recognition (OCR) is performed via EasyOCR or Tesseract depending on text clarity.

iv. **Intent Classification:** A heuristic keyword-based scorer classifies the transaction intent into primary categories: PAYMENT_IN, ORDER_IN, or COMPLAINT.

v. **AI Structuring:** The raw OCR text is processed by a Large Language Model (Qwen 2.5-72B), which extracts specific values: Item Description, Amount (MYR), Transaction Date, and a Buyer Sentiment Score (1–10).

vi. **Persistence:** The structured data is stored in the Convex real-time database, instantly reflecting in the user's dashboard.

2. **VernStudio (Snap & Sell)**



This module enables rural MSMEs to generate professional, bilingual e-commerce listings by uploading a product photo and providing a voice description in vernacular dialects (e.g., Iban, Foochow, Sarawak Malay).

Execution Pipeline:

i. **Multimodal Input:** The user captures a product photo and records a short audio description in the browser (via the MediaRecorder API).

ii. **Audio Conversion:** The audio (WebM) is converted to a standardized 16kHz WAV format using FFmpeg on the backend.

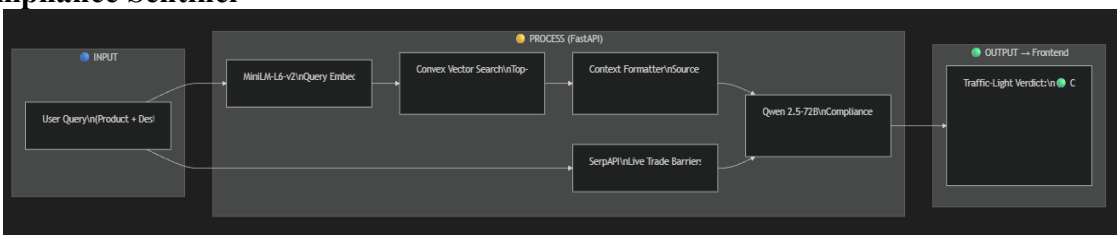
iii. **Speech-to-Text (ASR):** The mesolitica/whisper-50TPS-VQ-32k-large-v3-turbo model natively transcribes the vernacular audio into English/Malay text.

iv. **Image Processing:** The rembg library (U^2-Net architecture) strips the background from the product photo and composites the subject onto a clean white background.

v. **Listing Generation:** The Kimi-K2.5 multimodal model processes both the cleaned product image and the audio transcript to generate a highly optimized bilingual listing (English and Mandarin Chinese), including SEO tags.

vi. **Storage:** The enhanced image is uploaded to Convex file storage, and the resulting text is returned to the user interface.

3. **Compliance Sentinel**



This module utilizes a Retrieval-Augmented Generation (RAG) framework to provide immediate, contextually accurate compliance guidance based on the ASEAN Digital Economy Framework Agreement (DEFA).

Execution Pipeline:

i. **User Query:** The user inputs the product type and target destination country.

ii. **Vector Embeddings:** The query is embedded into a 384-dimensional vector space using the all-MiniLM-L6-v2 transformer model.

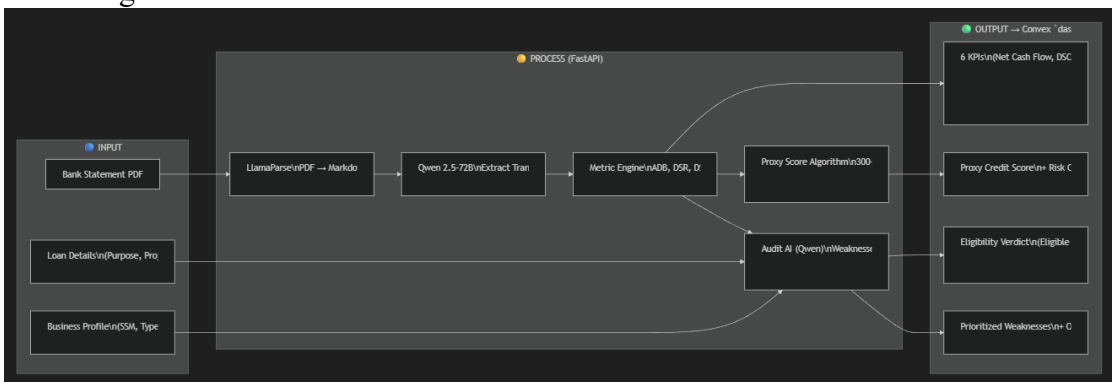
iii. **Similarity Search:** A vector search is executed against the defa_documents index in Convex, retrieving the top 5 most highly correlated regulatory clauses.

iv. **Context Formatting:** The retrieved regulatory documents are formatted into a context payload, prioritizing relevance scores and metadata.

v. **Rule Synthesis:** The Qwen 2.5-72B instruction model synthesizes the context and user query to produce a definitive "traffic-light" verdict (□ Clear, □ Conditional, ● Restricted), along with required documentation and estimated HS codes.

vi. **Delivery:** The structured JSON response is visualized as an actionable checklist on the frontend.

4. Pre-Screening Dashboard



This module bridges the gap between informal ledger tracking and formal institutional financing by evaluating uploaded bank statements alongside recorded shadow-ledger patterns to predict eligibility.

Execution Pipeline:

1. **Document Parsing:** The user uploads 3–12 months of bank statement PDFs, which are converted to structured markdown via LlamaParse, or passed as images to a vision model (Qwen-VL) if structural parsing fails.
2. **Metric Calculation:** The backend calculates critical financial ratios, including Average Daily Balance (ADB), Debt Service Ratio (DSR), volatility index, and expense ratios.
3. **AI Categorization (Qwen 2.5):** Unstructured transaction lines are categorized into revenue, fixed expenses, variable expenses, or debt repayments to formulate an accurate DSCR.
4. **Proxy Credit Scoring:** An algorithmic proxy score (spanning 300 to 850) is generated based on a weighted sum of asset coverage, revenue consistency, and ledger performance, acting as an alternative to traditional credit bureaus like CCRIS/CTOS.

5. **Macro Fusion:** The insights are fused with macro-economic ASEAN data (e.g., transit costs, logistics friction) to generate an actionable strategic summary and determine immediate financing eligibility.

4.0 Results

1. Architecture & Database Overhaul

- **Convex Migration & Normalization:** We successfully restructured the Convex database. We decoupled the monolithic frontend profiles table to segregate authentication data from intensive business intelligence metrics, yielding a standalone dashboard table. This greatly improved indexing strategy and reduced bloated queries.
- **Data Flow Standardization:** We established clear Input-Process-Output flows through custom data flow architectures for four core AI products:
    - **Shadow Ledger:** Orchestrating Pua Kumbu motif OCR logic to convert raw receipts into structured transaction metadata.
    - **Pre-Screening Dashboard:** Synthesizing unstructured data through the Qwen-3 model to output a robust Proxy Credit Score (300-850).
    - **Compliance Sentinel:** A regulatory map translating user trade intent into real-time custom tariffs and compliance metrics across ASEAN nations.
    - **VernStudio:** Converting snapshot data into vertical fast-ads tailored to regional tastes.

2. Robust Backend Processing

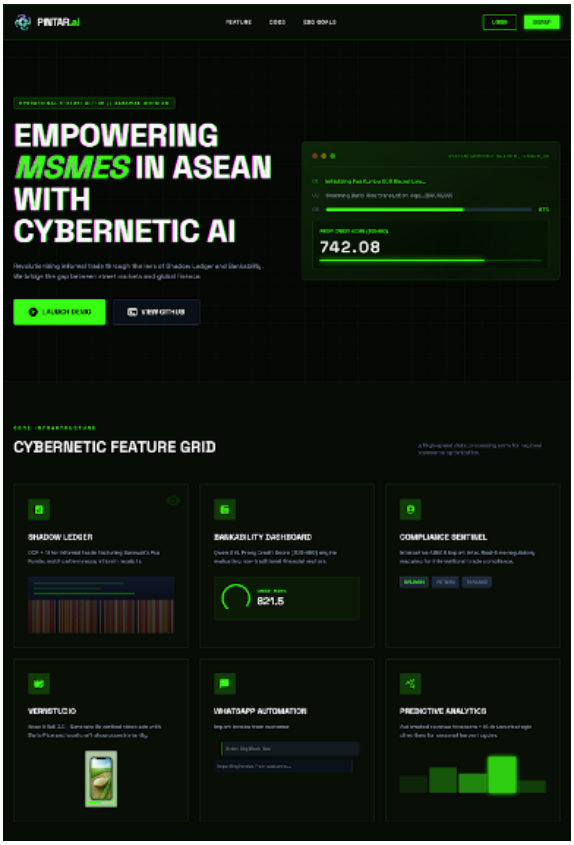
- **AI Integration Validation:** Deployed robust logic layers via FastAPI/Python to interact safely with Clerk's OAuth token system and process proxy score evaluations securely in the background without affecting frontend performance.
- **Migration scripts:** Deployed server-side Convex scripts to automate the one-time porting of existing profiles-based bankability metrics securely into the newly provisioned dashboard tables without losing legacy user credit scores.

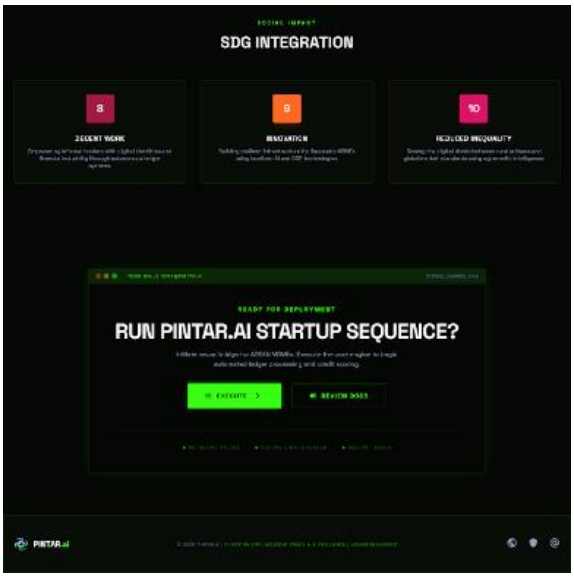
3. Frontend & Design Implementations

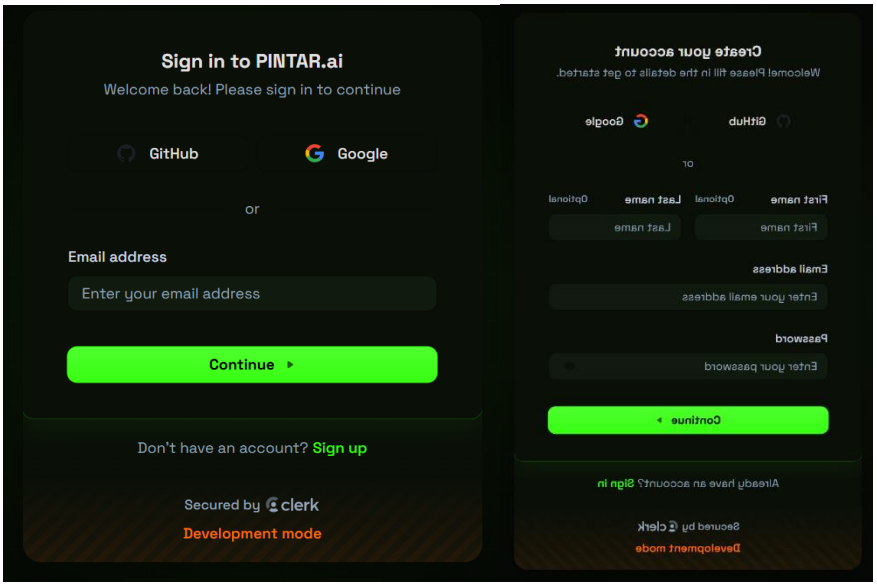
- **Next.js & Stitch Integration:** Migrated multiple raw HTML UI designs exported from Stitch into complex interactive Next.js 15 React components, completely preserving the *Cybernetic Feature Grid* cyberpunk aesthetics.
- **Premium Styling Transformation:** Converted generic Tailwind implementations into a cohesive Tailwind v4 Design System, routing custom Stitch color spaces (e.g., *primary*, *obsidian*, *neon-glow*) into precise arbitrary classes so the UI remained pristine—resulting in the final **Landing Page 2.0** mirroring the precise layout, spacing, margin, and typography ("Space Grotesk") of the original high-fidelity prototype.

5.0 Prototype Showcase

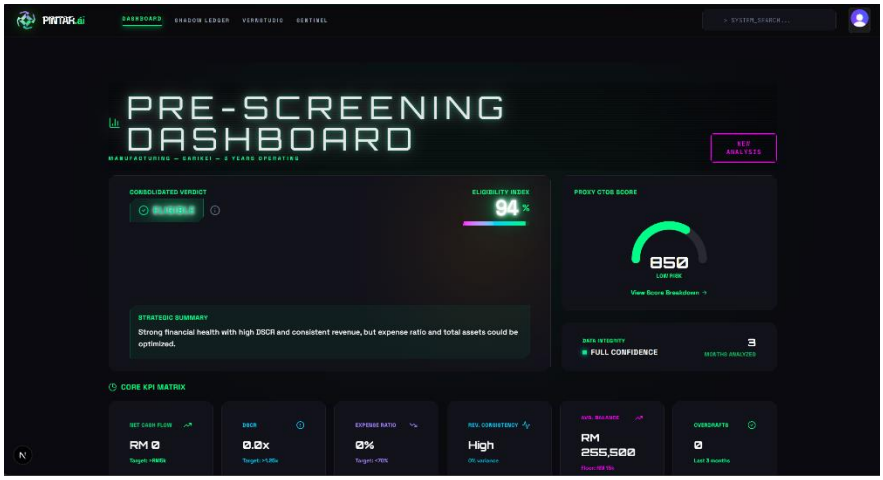
1. Landing Page



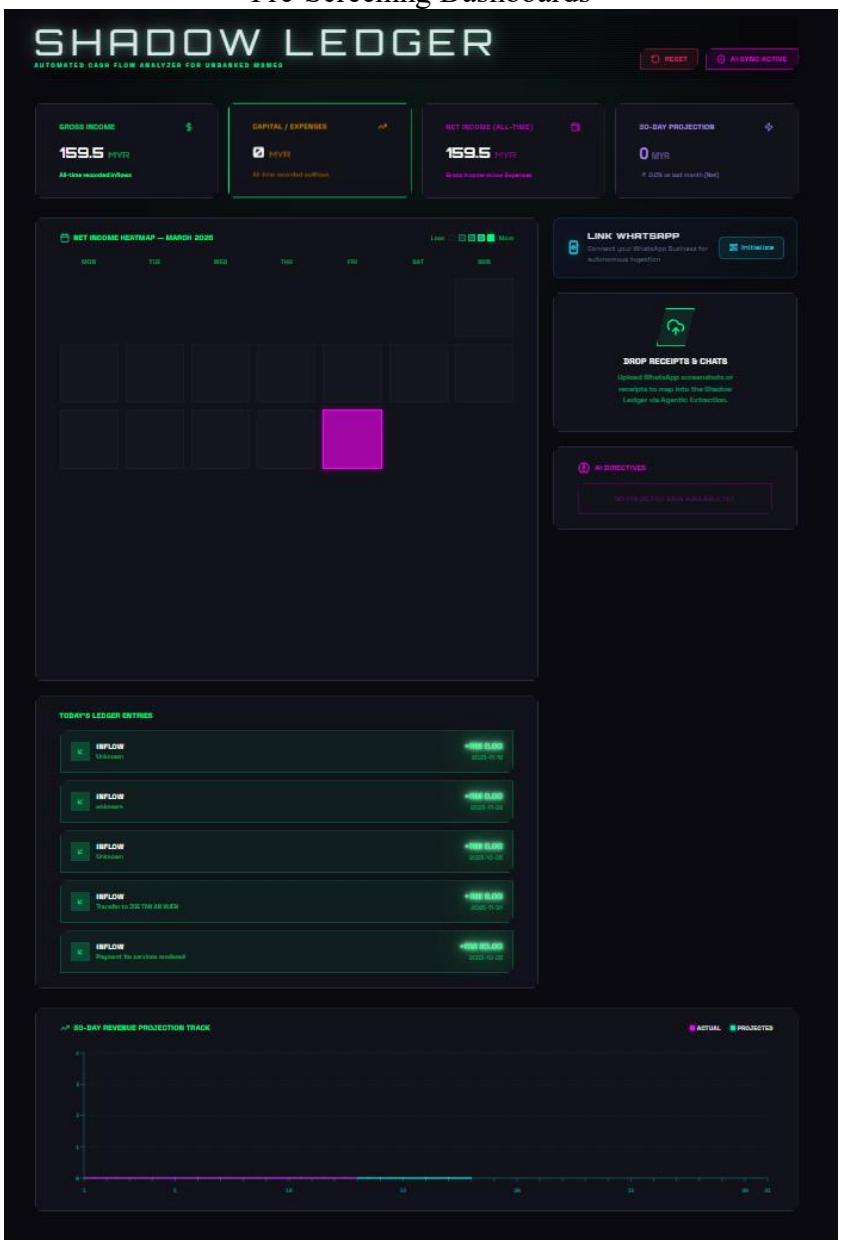




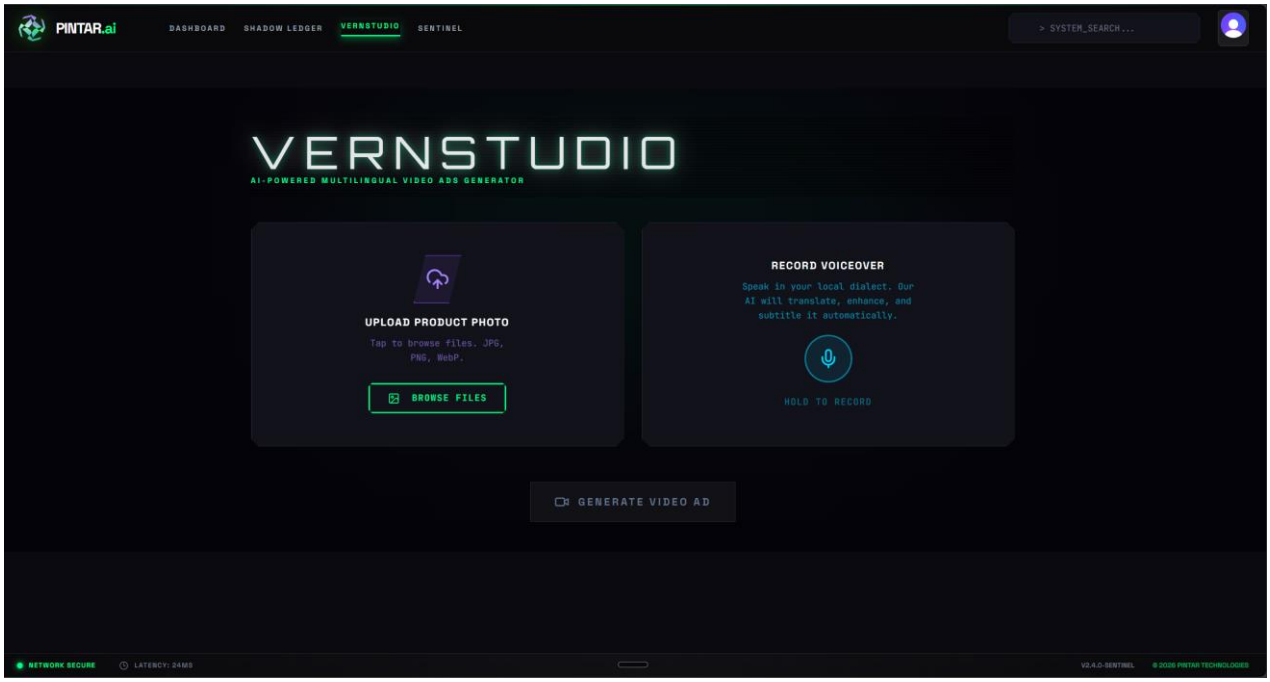
User Registration & Login



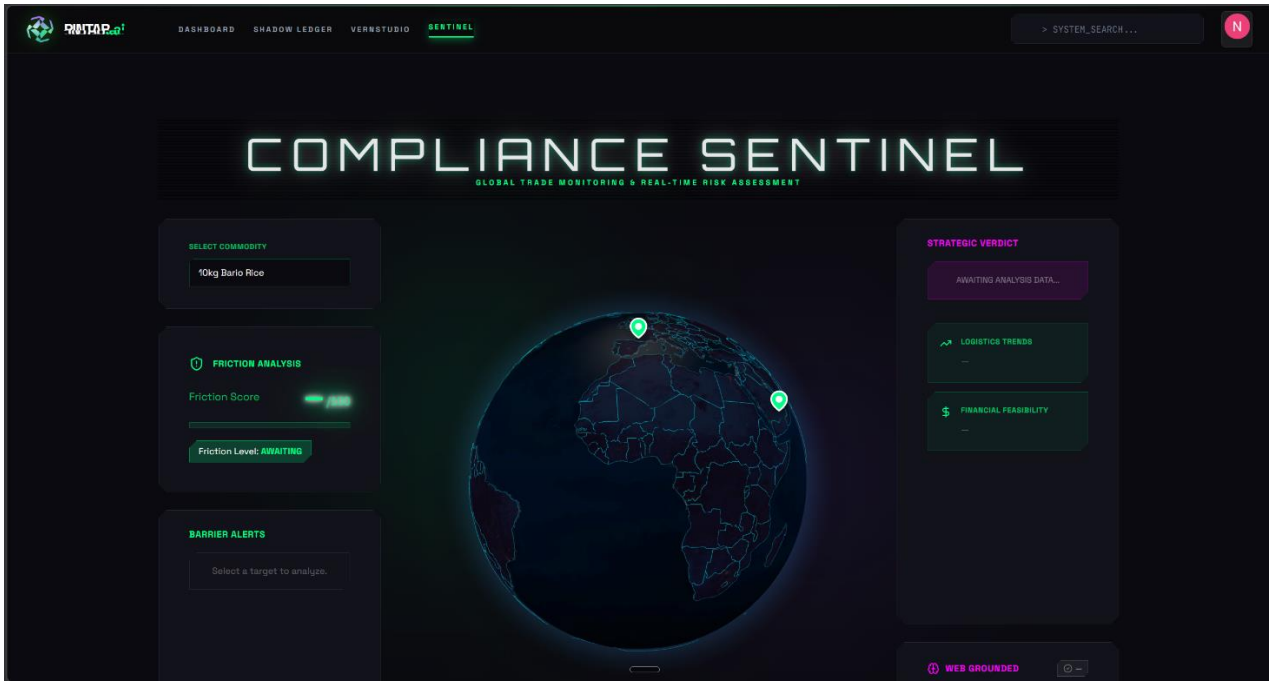
Pre-Screening Dashboards



Shadow Ledger



VernStudio



Compliance Sentinel

6.0 AI Acknowledgement (Mandatory)

PINTAR.ai is heavily powered by a plethora of LLMs and GenAI to accomplish and demonstrate the methodologies mentioned earlier. In terms of LLM that offers interpretations of instructions and evaluates ideas, PINTAR.ai uses *Qwen2.5 VL 32B Instruct* abundantly over predictive analytics dashboard and compliance checker. With its fast response and sufficient computational power, this empowers PINTAR.ai to address valuable outcomes accurately to the users' end. Meanwhile, in order to precisely evaluating and extracting semantics and data from documents such as bank statements PDF or images like receipts, PINTAR.ai uses *Kimi K2.5, Qwen 3 VL, EasyOCR (Optical Character Recognition)*. Meanwhile, PINTAR.ai use *maalysia-ai*'s fine-tuned *Whisper Large V3 Turbo* by *OpenAI* to effectively analyse audio captured in Malaysian languages (Iban, Bahasa Sarawak, Chinese Dialects, Mixed Languages). For GenAI, PINTAR.ai is powered by *LTX-2* by *Lightricks*, which is a diffuser joint audio-video generator to produce cinematic level of short advertisements videos. These LLMs and GenAI are **orchestrated** for most of the features in PINTAR.ai to perform "**agentically**" using *LangGraph*. On behalf of the system consistency and long-term functionality, the

development of PINTAR.ai uses self-fine-tuned open-source models instead of dominant industrial models.

Throughout the development of PINTAR.ai, a wide array of AI tools was utilised to resolve an ample amount challenges and errors with accurate diagnosis and guidance:

AI Tools	Usage
Google AI Studio	Prototyping/Seeking improvements on core features of PINTAR.ai
Google Stitch	Prototyping UI/UX Templates that were used as reference to design PINTAR.ai
Grok	Research Tool to verify existing solutions and further ascertain status quo of MSMEs in ASEAN
ChatGPT	Brainstorming ideas and system development blueprint, structure and technical stack
Claude AI	Debugging tool and tech assistant to explore technical setup on development away from "comfort zones"

7.0 Conclusion

The **PINTAR.ai** project has successfully graduated from theoretical designs and unoptimized data storage into a robust, scalable, and highly stylized MVP. By isolating the database architecture, ensuring type-safe API communication between Next.js, FastAPI, and Convex, and implementing a pixel-perfect, highly engaging frontend, PINTAR.ai is now structurally prepared to scale.

The platform succeeds in its technical promise: it now possesses the necessary data pipelines to securely ingest raw MSME trade data, the backend intelligence to evaluate that data dynamically into structured proxy credit scores, and the frontend capability to present this in an intuitive, immersive dashboard. PINTAR.ai stands as an inclusive growth engine — firmly ready for ASEAN deployment.

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